COMPUTER NETWORKS FINAL EXAMINATION (2022/2023) – Moad A

Gadi Taubenfeld

Question 1

A circuit-switched network - True
The Alternating Bit protocol - True
In TCP - True
In the DHCP - True
A user will get - False
A NAT may change - False
Consider the consensus - True
A bridge may update – False
In the Ethernet protocol - True
ARP queries – False

Question 2

Suppose an ISP – 32
Hosts A and Host B – 451
Consider a client – 11
The question refers – 1/128

Question 3

1. 68 msec

2.

A	40
B	250
C	40
D	40
E	250
F	50
G	290
H	55
I	330
J	330
K	50

3.

The number of edges that become core edges is: **7**
The level of the fragment is: **3**

Question 4
1.

Step	Source IP	Destination IP	Source Port	Destination Port
1 – from host	198.3.3.3	222.1.1.9	**2000**	500
2 – from NATted router	**55.2.1.3**	**222.1.1.9**	7000	500
3 -- from server	222.1.1.9	55.2.1.3	500	**7000**
4 -- from NATted router	222.1.1.9	**198.3.3.3**	**500**	2000

Internet side	LAN-side
55.2.1.3, 7000	198.3.3.3, 2000

2.
Answers for 1 and 2:

Network with four switches and one router		
Point	**Number of ARP packets**	**Number of IP packets**
(1)	**4**	**6**
(2)	**4**	**6**
(3)	**4**	**2**
(4)	**8**	**4**
(5)	**6**	**2**
(6)	**6**	**2**

Answers for 3 and 4:

Network with five switches		
Point	**Number of ARP packets**	**Number of IP packets**
(1)	**12**	**6**
(2)	**12**	**6**
(3)	**8**	**2**
(4)	**10**	**4**
(5)	**8**	**2**
(6)	**8**	**2**

Answers for 5 and 6:

Network with four hubs and one router		
Point	**Number of ARP packets**	**Number of IP packets**
(4)	**8**	**4**
(5)	**8**	**4**
(6)	**8**	**4**